

A GLIMPSE OF COMPUTABILITY THEORY IN FRACTAL GEOMETRY

ZACH ARCHIBALD

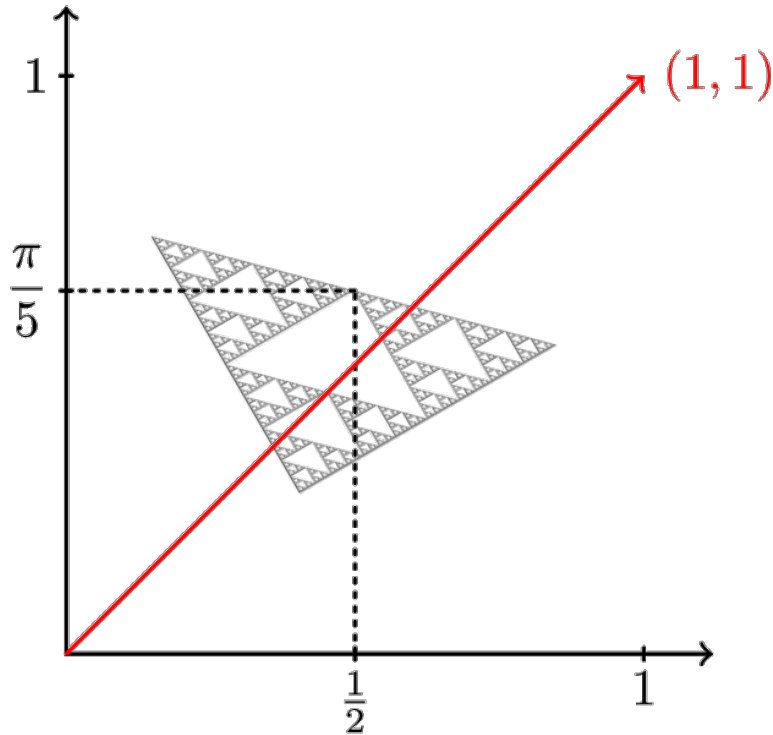
INTRODUCTION

Let S be the Sierpinski gasket, and $\mathcal{W} \curvearrowright \mathbb{R}^2$ be an iterated function system with attractor $A_{\mathcal{W}} = \Psi(S)$, where

$$\Psi : \mathbb{R}^2 \curvearrowright \text{ by } \Psi \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} \frac{1}{4} & 0 \\ 0 & \frac{1}{4} \end{pmatrix}$$

is the affine transformation that rotates by 30° , scales each coordinate by $\frac{1}{2}$, translates x positively by $\frac{1}{4}$ and translates y positively by $\frac{\pi}{5}$.

Consider alongside $A_{\mathcal{W}}$ the line segment L from $\langle 0, 0 \rangle$ to $\langle 1, 1 \rangle$.



We seek to determine if L intersects $A_{\mathcal{W}}$.

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One reasonable thought is to check each iterate of $\mathcal{H}(\mathcal{W})$, the image of Hutchinson operator, for intersection. If we find an iterate whose image does not intersect L , we know $A_{\mathcal{W}}$ does not intersect L , since $A_{\mathcal{W}} \subset \mathcal{H}^i(\mathcal{W})$ for any $i \in \mathbb{N}$.

However, so long as L continues to intersect each $\mathcal{H}^i(\mathcal{W})$, we can not conclude anything about L 's intersection with $A_{\mathcal{W}}$; it may be the case that a further iterate of the Hutchinson operator certifies disjointedness, but it may just as well be the case that there is in fact an intersection between L and $A_{\mathcal{W}}$, whereby this procedure would continue on forever, checking each further iterate of the Hutchinson operator for intersection.

Its clear that this procedure is not sufficient to determine if the provided intersection occurs. We begin to question whether or not such a procedure exists.

ABSTRACTION AND FORMALIZATION

To solve this problem, we instead consider (in place of the Sierpinski gasket similitude) an arbitrary iterated function system acting on the unit square. To formalize the question, we require some definitions from computability theory.

Definition. A decision problem is a problem which has a yes or no answer.

Example. Given a list of real numbers N and another real number x , the problem of determining whether N contains x is one of decision, as x either is or is not contained in N .

Definition. A decision problem is considered decidable if and only if there exists an algorithm that solves any instance of the problem in finite time.

Example. The decision problem of the previous example is decidable, as the following algorithm provides the solution for any input in finite time:

For each $n \in N$ check for equality with x . If equality is found, return yes. If every $n \in N$ has been checked and equality has not been found, return no.

With this terminology, our guiding question can be articulated: Is it decidable to test if the attractor of a given iterated function system acting on $\mathbb{R}^2 \cap [0, 1]^2$ intersects L ?

TOOLS AND DIRECTION

Definition. Let P and Q be distinct problems. We say P *reduces* to Q if there exist algorithms Φ and Ψ such that

- Φ converts P_i , an instance of problem P , into Q_i , an instance of problem Q
- Ψ converts the solution of Q_i into the solution of P_i

Example. The problem of finding the k th largest element of a list of numbers reduces to the problem of sorting by the following reduction: Let Φ be the identity algorithm (which returns its input), and Ψ return the first element in the sorted list. Informally, to find the largest element of a list, we can sort the list from greatest to least and return the k th element of the sorted list.

A common way to identify the decidability of a problem is via reduction.

Theorem 1. If we can reduce a problem P to a decidable problem, P is decidable. Symmetrically, if we can reduce an undecidable problem to P , it must be the case that P is undecidable; otherwise, our undecidable problem could be decided through this reduction—a contradiction.

As the classical theory of computation builds upon formal language theory, there is a considerable collection of problems with established decidability results in the latter area. To engage with these problems, we must first introduce a few definitions.

Definition. Let Σ be a nonempty, finite set of characters (an alphabet). We denote the finite strings (or words) made from Σ as Σ^* , the infinite strings made from Σ as Σ^ω , and finally $\Sigma^\infty = \Sigma^* \cup \Sigma^\omega$. For any $w \in \Sigma^\infty$, we denote the cardinal number of symbols in w with $|w|$. The binary operation of Σ^* 's free monoid is referred to as concatenation, with identity ϵ , the empty string. Concatenation is further defined on $\Sigma^* \times \Sigma^\infty$ —we allow both finite and infinite strings to be left concatenated by finite strings.

Equipped with these definitions, we present an undecidable problem:

Definition. I-PCP: Given lists A and B of k strings each from Σ^* , say

$$A = a_1, a_2, \dots, a_k \quad \text{and} \quad B = b_1, b_2, \dots, b_k,$$

does there exist a sequence of integers $i_\bullet = \{i_1, i_2, \dots\}$ with $1 \leq i_\bullet \leq k$ such that

$$a_{i_1} a_{i_2} \dots = b_{i_1} b_{i_2} \dots?$$

The sequence $i_\bullet$ is called a *solution* of this instance of I-PCP.

We seek a reduction from I-PCP to the IFS attractor-line intersection problem. The key is to interpret these language structures in I-PCP as addresses in the code space of some IFS acting on $[0, 1]^2$.

FINDING A REDUCTION FROM I-PCP TO THE IFS ATTRACTOR-LINE INTERSECTION PROBLEM

Notice that I-PCP is a problem of strings, while the IFS attractor-line intersection problem is one of numbers in the unit square. As such, it is reasonable to assume any reduction between these problems would require a sensible map from Σ^∞ to $[0, 1]$.

Definition. For any $\Sigma = \{1, 2, \dots, k\}$, let $I : \Sigma^\infty \rightarrow [0, 1]$ denote the Dube interpretation function. We define the mapping in the following way [1]:

For the empty string ϵ , let $I(\epsilon) = 0$. A non-empty string $w \in \Sigma^*$ (note that 0 is not in $\Sigma = \{1, \dots, k\}$) is interpreted as a rational number by putting the radix point to the left and interpreting the string in base $k + 1$ notation. Elements of Σ^ω are interpreted in a similar fashion.

Note: While this definition restricts Σ , for any finite alphabet, we can work instead with an enumeration of its elements (which conforms to this restriction).

Example. Let $\Sigma = \{1, 2, 3, 4\}$. Noting that the interpretations are represented in base 5 notation, a few strings' interpretations follow:

- $I(\epsilon) = 0$
- $I(1234) = 0.1234$
- $I(111\dots) = 0.111\dots$

The Dube interpretation function provides a way to concatenate strings via affine transformations on their interpretations:

Definition. Let $\Sigma = \{1, 2, \dots, k\}$ and let $I : \Sigma^\infty \rightarrow [0, 1]$ be the Dube interpretation function. For every string $w \in \Sigma^*$, define a one-dimensional affine transformation $f^w : [0, 1] \rightarrow [0, 1]$ with

$$f^w(x) = \frac{1}{(k+1)^{|w|}}x + I(w).$$

Theorem 2. The following hold:

- (1) I is non-surjective and its restriction to Σ^ω is injective.

Proof. Fix $k \geq 1$ and $\Sigma = \{1, 2, \dots, k\}$. It is clear from the definition of I that $I^{-1}(0.101) = \emptyset$ since $0 \notin \Sigma$, providing by certificate the non-surjectivity of I .

Retaining the prior selections, fix $I(\omega), I(\gamma) \in I(\Sigma^\omega)$ such that $I(\omega) = I(\gamma)$. $I(\omega) = I(\gamma)$ has either one base $k+1$ representation, or two base $k+1$ representations such that one ends in repeated k s and the other ends in repeated 0s. Since $0 \notin \Sigma$, we know that $I(\omega)$ and $I(\gamma)$ cannot be of the latter representation, and thus they have the same base $k+1$ representations. Suppose toward contradiction that $\omega \neq \gamma$. Then for some $i \in \mathbb{N}$, $\omega_i \neq \gamma_i$. It is clear from the definition of I that this would imply that the i th digits of $I(\omega), I(\gamma)$ disagree, contradicting our assumption. We have shown that I restricted to Σ^ω injects. $\square$

- (2) Let $w \in \Sigma^*$ and let $x \in \Sigma^\infty$. Then $I(wx) = f^w(I(x))$.

Proof.

$$\begin{aligned} I(wx) &= 0.w_1\dots w_{|w|}x_1x_2\dots \\ &= 0.0\dots 0x_1x_2\dots + 0.w_1\dots w_{|w|} \\ &= \frac{1}{(k+1)^{|w|}}I(x) + I(w) \\ &= f^w(I(x)) \end{aligned}$$

$\square$

- (3) Let $u_1, u_2, \dots, u_i \in \Sigma^*$ where $i \geq 1$. Then

$$I(u_1u_2\dots u_i) = f^{u_1}(f^{u_2}(\dots(f^{u_i}(0))\dots)).$$

Proof. By definition of f^{u_i} it is clear that $f^{u_i}(0) = I(u_i)$. Utilizing (2), we have

$$\begin{aligned} I(u_1 u_2 \dots u_i) &= f^{u_1}(I(u_2 u_3 \dots u_i)) \\ &= f^{u_1}(f^{u_2}(I(u_3 u_4 \dots u_i))) \\ &\vdots \\ &= f^{u_1}(f^{u_2}(\dots (f^{u_{i-1}}(I(u_i)) \dots)) \\ &= f^{u_1}(f^{u_2}(\dots (f^{u_i}(I(\epsilon)) \dots)) \\ &= f^{u_1}(f^{u_2}(\dots (f^{u_i}(0)) \dots)) \end{aligned}$$

□

(4) Let $u_1, u_2, \dots \in \Sigma^*$. Then

$$I(u_1 u_2 \dots) = \lim_{i \rightarrow \infty} f^{u_1}(f^{u_2}(\dots (f^{u_i}(x)) \dots))$$

where x is any real number.

Proof. As in (3), repeated application of (2) provides

$$I(u_1 u_2 \dots) = \lim_{i \rightarrow \infty} f^{u_1}(f^{u_2}(\dots (f^{u_i}(0)) \dots))$$

Furthermore, note that each f^{u_i} is contraction with contractivity factor $\frac{1}{(k+1)^{|u_i|}}$.

Let $s = \max_{i \in \mathbb{N}} \left\{ \frac{1}{(k+1)^{|u_i|}} \right\}$. We have for any $i \in \mathbb{N}$ and any $x \in \mathbb{R}$ that

$$|f^{u_1}(f^{u_2}(\dots (f^{u_i}(0)) \dots)) - f^{u_1}(f^{u_2}(\dots (f^{u_i}(x)) \dots))| \leq s^{|u_i|}|x|$$

as $i \rightarrow \infty$, $s^{|u_i|}|x| \rightarrow 0$; we conclude that these expressions equate within the limit—that for any $x \in \mathbb{R}$,

$$I(u_1 u_2 \dots) = \lim_{i \rightarrow \infty} f^{u_1}(f^{u_2}(\dots (f^{u_i}(x)) \dots))$$

□

(5) Let $u_1, u_2, \dots, u_N \in \Sigma^*$ and let $R = \{u_{i_1} u_{i_2} u_{i_3} \dots : 1 \leq i_\bullet \leq N\} \subseteq \Sigma^\omega$ for some $N \geq 1$. Let A be the attractor of the IFS $\{[0, 1]; f^{u_1}, f^{u_2}, \dots, f^{u_N}\}$. Then $I : R \rightarrow A$ is a bijection.

Proof. Recall that the attractor of the IFS is

$$A = \bigcup_{w \in R} \lim_{n \rightarrow \infty} f^{w_1}(f^{w_2}(\dots (f^{w_n}(x)) \dots)),$$

where $w = w_1 w_2 w_3 \dots$, $w_i \in \{u_1, u_2, \dots, u_N\}$ for all i , and x is any real number. That is,

$$A = \bigcup_{w \in R} I(w_1 w_2 w_3 \dots) = \bigcup_{w \in R} I(w).$$

Thus $A = I(R)$ and therefore $I : R \rightarrow A$ is surjective. And since I is injective on Σ^∞ , we see that $I : R \rightarrow A$ is bijective. $\square$

This concludes the theorem.

We now extend to the previous theorem to 2 dimensions, interpreting pairs of strings as doubles. In particular, $I : \Sigma^\infty \times \Sigma^\infty \rightarrow [0, 1]^2$ by $I(\omega, \eta) = \langle I(\omega), I(\eta) \rangle$.

Theorem 3. Let $\Sigma = \{1, 2, \dots, k\}$ and let $I : \Sigma^\infty \times \Sigma^\infty \rightarrow [0, 1]^2$ be the interpretation function defined above.

Let $(a_1, b_1), (a_2, b_2), (a_3, b_3), \dots \in \Sigma^* \times \Sigma^*$. Let $g_i : [0, 1]^2 \rightarrow [0, 1]^2$ by

$$g_i \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} (k+1)^{-|a_i|}x + I(a_i) \\ (k+1)^{-|b_i|}y + I(b_i) \end{pmatrix}$$

Then the following hold:

- (1) I is non-surjective and its restriction to $\Sigma^* \times \Sigma^*$ is injective.
- (2) Let $(r, s) \in \Sigma^\infty \times \Sigma^\infty$. Then for all $i \geq 1$,

$$I((a_i, b_i), (r, s)) = g_i(I(r, s)).$$

- (3) For all $i \geq 0$,

$$\begin{pmatrix} a_1 a_2 \dots a_i \\ b_1 b_2 \dots b_i \end{pmatrix} = g_i(g_{i-1}(\dots (g_1(0)) \dots)),$$

where 0 is the zero vector.

- (4) Let z be any point in $[0, 1]^2$. Then

$$\begin{pmatrix} a_1 a_2 \dots \\ b_1 b_2 \dots \end{pmatrix} = \lim g_1(g_2(\dots (g_i(z)) \dots)).$$

- (5) Let $R = \{(a_{i_1} a_{i_2} \dots, b_{i_1} b_{i_2} \dots) : 1 \leq i_\bullet \leq k\}$, and let $A_{\mathcal{W}}$ be the attractor of the IFS $\{[0, 1]^2; g_1, g_2, \dots, g_N\}$. Then $I : R \rightarrow A_{\mathcal{W}}$ is a component wise bijection. That is, $I : \pi_1(R) \rightarrow \pi_1(A_{\mathcal{W}})$ is a bijection, and so is $I : \pi_2(R) \rightarrow \pi_2(A_{\mathcal{W}})$

We forgo the proofs of these claims, as they follow directly from Theorem 3 and the properties of the Cartesian product. This concludes the theorem.

Theorem 3 item 5 provides a reduction from I-PCP to the IFS attractor-line intersection problem.

REDUCING I-PCP TO IFS ATTRACTOR-LINE INTERSECTION PROBLEM

Theorem 4. For any instance of I-PCP (for any Σ , $k \in \mathbb{N}$, $A = a_1, a_2, \dots, a_k$ and $B = b_1, b_2, \dots, b_k$ such that $a_\bullet, b_\bullet \in \Sigma^*$), let $\mathcal{W} \curvearrowright [0, 1]^2$ by

$$\mathcal{W} = \left\{ f_j \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} (k+1)^{-|a_j|}x + I(a_j) \\ (k+1)^{-|b_j|}y + I(b_j) \end{pmatrix} : 1 \leq j \leq k \right\}$$

There exists a solution to this instance of I-PCP if and only if $A_{\mathcal{W}} \subset [0, 1]^2$ intersects the line segment from $\langle 0, 0 \rangle$ to $\langle 1, 1 \rangle$.

Proof. Fix an instance of I-PCP. That is, fix Σ , $k \in \mathbb{N}$, $A = \{x_1, x_2, \dots, x_k\}$ and $B = \{y_1, y_2, \dots, y_k\}$ such that $x_\bullet, y_\bullet \in \Sigma^*$.

There exists a solution to this instance of I-PCP if and only if there exists an $\{i_\bullet\}$ such that $a_{i_1}a_{i_2}\dots = b_{i_1}b_{i_2}\dots$ if and only if $\{(a_{i_1}a_{i_2}\dots, b_{i_1}b_{i_2}\dots) : 1 \leq i_\bullet \leq k\}$ has a diagonal point.

By theorem 3 item 5, $\{(a_{i_1}a_{i_2}\dots, b_{i_1}b_{i_2}\dots) : 1 \leq i_\bullet \leq k\}$ bijects component-wise to $A_{\mathcal{W}}$ through I . This implies $\{(a_{i_1}a_{i_2}\dots, b_{i_1}b_{i_2}\dots) : 1 \leq i_\bullet \leq k\}$ has a diagonal point if and only if $A_{\mathcal{W}}$ has a diagonal point.

Finally, $A_{\mathcal{W}} \subset [0, 1]^2$ has a diagonal point if and only if it intersects the line segment from $\langle 0, 0 \rangle$ to $\langle 1, 1 \rangle$.

Chaining these biconditionals, we have that there exists a solution to this instance of I-PCP if and only if $A_{\mathcal{W}} \subset [0, 1]^2$ intersects the line segment from $\langle 0, 0 \rangle$ to $\langle 1, 1 \rangle$. □

Theorem 5. I-PCP reduces to the IFS attractor-line intersection problem.

Proof. For any instance of I-PCP, generating the IFS in theorem 4 associates an instance of the IFS attractor-line intersection problem to the instance of I-PCP. Given a solution to this instance of IFS attractor-line intersection problem, theorem 4 provides a solution to the instance of I-PCP. As a result, we have a reduction: Let Φ generate the $\mathcal{W}$ associated with the instance of I-PCP, and let Ψ return yes (I-PCP has a solution) if $A_{\mathcal{W}}$ intersects the line segment from $\langle 0, 0 \rangle$ to $\langle 1, 1 \rangle$, and return no otherwise. □

THE UNDECIDABILITY OF THE IFS ATTRACTOR-LINE INTERSECTION PROBLEM

Theorem 6. It is undecidable to test if the attractor of a given iterated function system acting on $\mathbb{R}^2 \cap [0, 1]^2$ intersects the line segment from $\langle 0, 0 \rangle$ to $\langle 1, 1 \rangle$.

Proof. Theorem 5 identifies a reduction from I-PCP to IFS Attractor-Line Intersection Problem. Since I-PCP is undecidable, Theorem 1 provides that the IFS Attractor-Line Intersection Problem is thus undecidable as well. □

REFERENCES

- [1] Simant Dube. Undecidable problems in fractal geometry. *Complex Systems*, 7(6):423–444, 1993.